NP-HARD & NP-COMPLETE PROBLEMS

Reduction

- We have two problems, A and B, and we know problem B is a P class problem.
- If problem A can be reduced, or converted to problem B, and this reduction takes a polynomial amount of time, then we can say that A is also a P class problem (A is reducible to B).

NP-HARD

- A problem is **NP-hard** if all problems in NP are polynomial time reducible to it, even though it may not be in NP itself.
- Any decision problem P_i is called NP-Hard if and only if every problem of NP is reducible to P_i in polynomial time.
- A problem is classified as NP-Hard when an algorithm for solving it can be translated to solve *any* NP problem.
- Then we can say, this problem is *at least* as hard as any NP problem, but it could be much harder or more complex.

  Example: Vertex Cover, Travelling Salesman Problem

NP-COMPLETE

- NP-Complete problems are problems that live in both the NP and NP-Hard classes.
- This means that NP-Complete problems can be verified in polynomial time and that any NP problem can be reduced to this problem in polynomial time.

- A problem is in the class NPC if it is in NP and is as hard as any problem in NP.
- If a polynomial time algorithm exists for any of these problems, all problems in NP would be polynomial time solvable.
- These problems are called NP-complete.

  Example: Determining whether a graph has a Hamiltonian cycle, Determining whether a Boolean formula is satisfiable, etc.

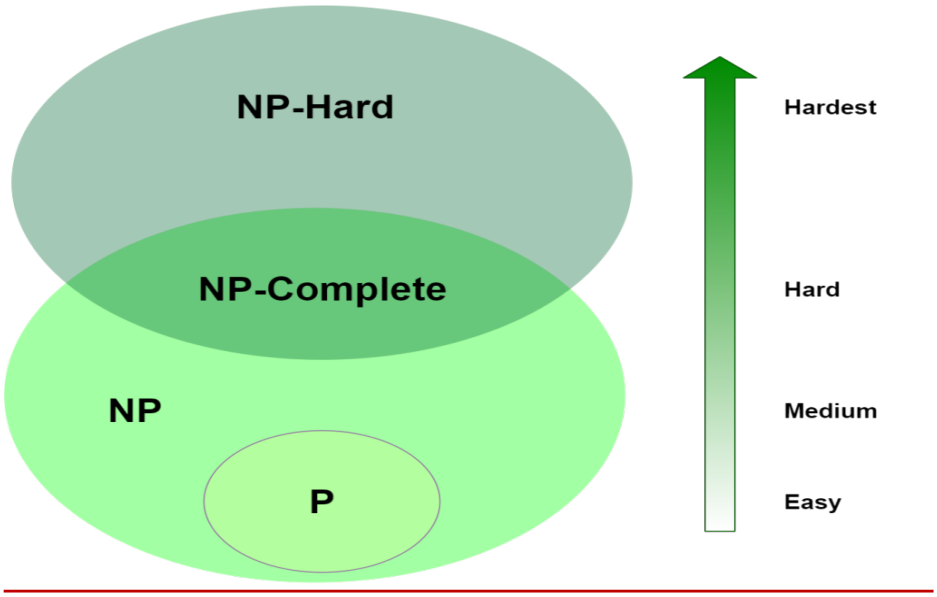
SUMMARY

P problems: quick to solve

NP problems: quick to verify but slow to solve

NP-Complete problems: quick to verify, slow to solve and can be reduced to any other NP-Complete problem

NP-Hard problems: slow to verify, slow to solve and can be reduced to any other NP problem



STUDY OF NP-COMPLETE PROBLEMS

